

Session 2: Spatial foundations, data acquisition and quality

Modeling the distribution of biodiversity under global climate change

Principles of Geographic Information Systems (GIS)

What is GIS? Why does it matter for modeling biodiversity?

Definition: a system to capture, store, manipulate, analyze, and present spatial or data.

Relevance in macroecology: mapping habitats, tracking species distributions under climate change, analyzing environmental patterns (e.g., temperature), conservation planning (MPAs).

Specific link to SDMs: supports the representation of species locations and environmental variables; crucial for data preparation, interpretation and visualization.

Representing Reality: Vector Data Models

Represents **discrete features with distinct boundaries**.

Primitives:

Points: x, y (,z) locations (e.g., occurrences, stations).

Lines: points connected by segments (e.g., coastlines, tracks, currents).

Polygons: closed sequence of segments representing areas (e.g., MPAs, islands).

Features are linked to an attribute table (e.g., species name, MPA size).

Can store spatial relationships (e.g., distances between sites).



Spatial information for *Caretta caretta*

Locations

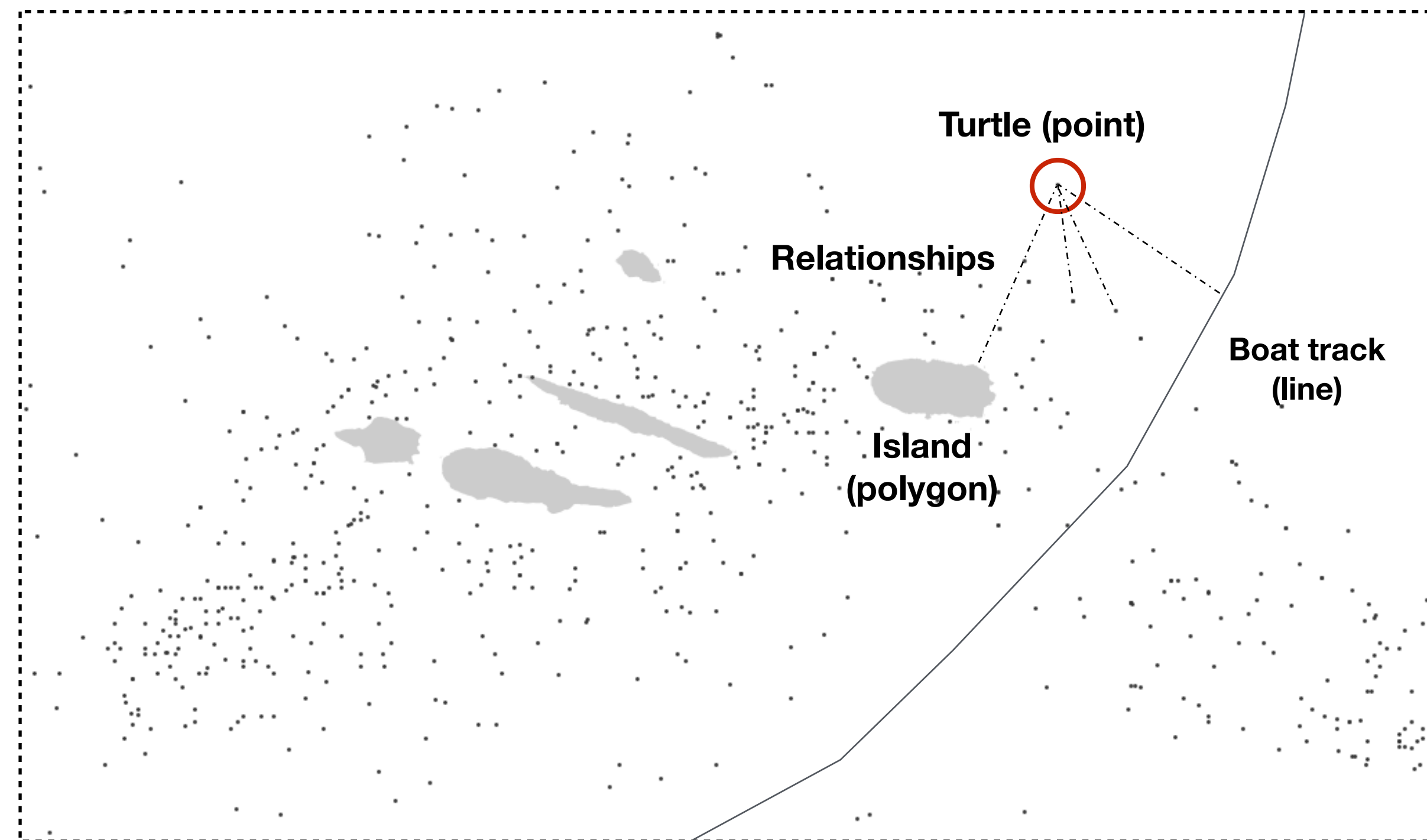
Attributes

Relationships

Lon	Lat	Species	Common name	Date	Depth	Behaviour	Observer	Distance to land
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	10/08/12	22	Swimming	John Doe	232
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	12	Swimming	John Doe	3454
3,18192	42,40479	Caretta caretta	Loggerhead sea turtle	14/08/11	1	Eating	Unknown	23
-4,86395	55,24295	Caretta caretta	Loggerhead sea turtle	01/01/04	0	Swimming	Unknown	2567
-5,89297	57,06977	Caretta caretta	Loggerhead sea turtle	03/03/12	0	Sleeping	John Doe	878
34,9082	32,55822	Caretta caretta	Loggerhead sea turtle	22/12/02	23	Sleeping	John Doe	777
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	14/08/11	12	Swimming	John Doe	6
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	2	Sleeping	John Doe	78

Vector data model

Points, lines and polygons (locations, and features with attributes)



The distribution of the Loggerhead sea turtle in the Azores Islands

Common file formats

Vector data are common as **shapefiles** (a standard since 1990).

.shp is the geometry of vector features

.dbf are the attributes of vector features

.shx helps the GIS Application to find features more quickly.

Vector data are migrating to simpler formats like **GeoPackage files**, which are an open, portable, self-describing, compact formats for spatial information.

.gpkg is the geometry and the attributes of vector features.

ID	SPECIES Y	LON	LAT	REF	
1	Sacchorhiza polyschides	1966	-5.600819	36.018775	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
2	Sacchorhiza polyschides	1993	-5.601822	36.019239	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
3	Sacchorhiza polyschides	1960	-5.653864	36.062617	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
4	Sacchorhiza polyschides	1965	-5.44705	36.116542	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
5	Sacchorhiza polyschides	1958	-5.353586	36.14075	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
6	Sacchorhiza polyschides	1965	-5.922481	36.190028	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
7	Sacchorhiza polyschides	1966	-6.087778	36.277683	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
8	Sacchorhiza polyschides	1983	-5.158386	36.422061	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
9	Sacchorhiza polyschides	1982	-4.717386	36.493706	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
10	Sacchorhiza polyschides	1982	-4.882289	36.510117	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
11	Sacchorhiza polyschides	1982	-4.622617	36.571558	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
12	Sacchorhiza polyschides	1966	-4.421261	36.721283	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
13	Sacchorhiza polyschides	1970	-8.936489	37.009864	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
14	Sacchorhiza polyschides	1963	-8.978044	37.023033	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
15	Sacchorhiza polyschides	1963	-8.667864	37.079811	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
16	Sacchorhiza polyschides	1963	-8.266667	37.083333	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
17	Sacchorhiza polyschides	1956	-8.45	37.083333	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
18	Sacchorhiza polyschides	1970	-8.247875	37.089072	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
19	Sacchorhiza polyschides	1970	-8.469325	37.098511	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
20	Sacchorhiza polyschides	1970	-8.896	37.184133	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
21	Sacchorhiza polyschides	1954	-8.770147	37.757611	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
22	Sacchorhiza polyschides	1970	-8.796936	37.906147	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
23	Sacchorhiza polyschides	1970	-9.216664	38.416647	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
24	Sacchorhiza polyschides	1979	-9.1176	38.43685	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
25	Sacchorhiza polyschides	1970	-9.101489	38.444469	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
26	Sacchorhiza polyschides	1989	-8.984847	38.469364	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
27	Sacchorhiza polyschides	1982	-8.990794	38.495931	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
28	Sacchorhiza polyschides	1970	-9.355975	38.689786	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.
29	Sacchorhiza polyschides	1989	-9.483333	38.7	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
30	Sacchorhiza polyschides	1951	-9.269286	38.702911	"JosÈ Luis Izquierdo; Tom.s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÙn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX.

Common file formats

Vector data are also available as **text delimited files** (e.g., .txt, .csv), delimited by a **Tab, a comma or semicolon**. Useful for point data and matrices (e.g., excel).

Representing Reality: Raster Data Models

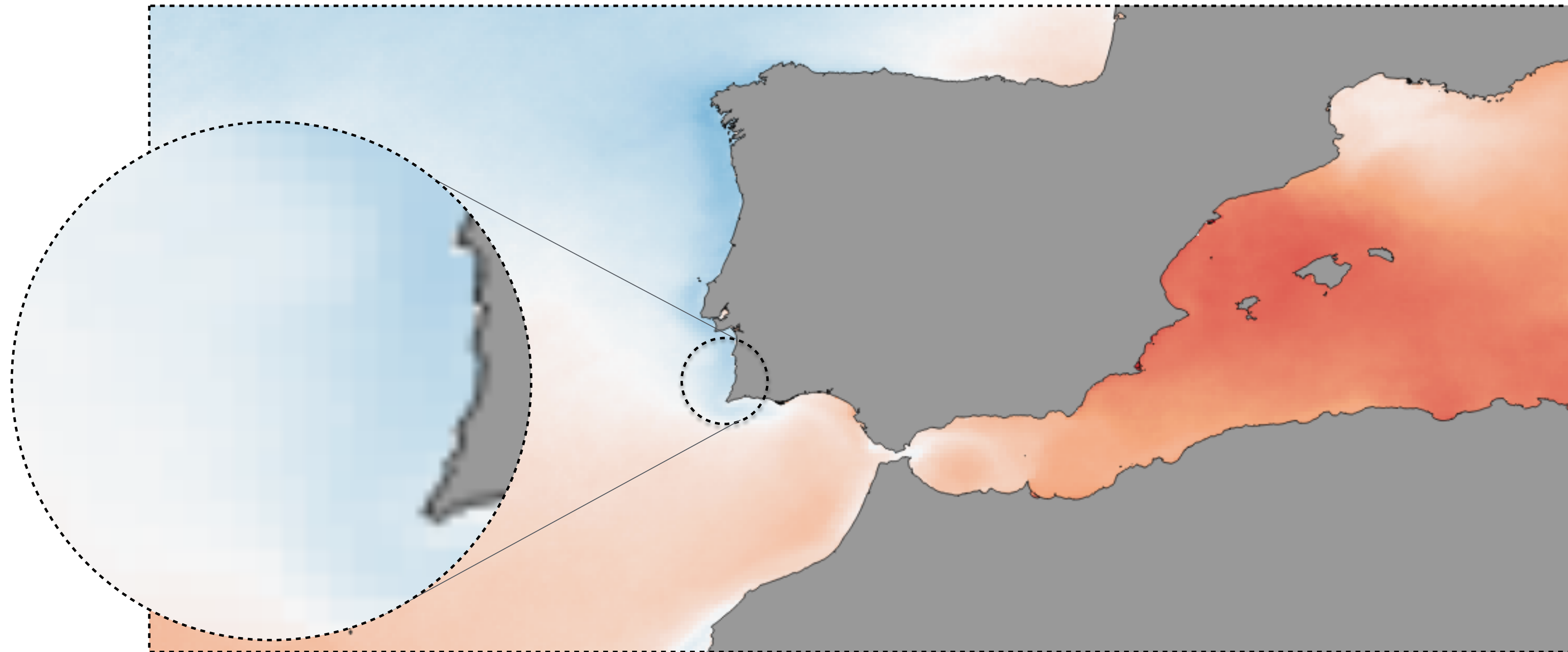
Composed of a **grid of cells** (pixels), where each cell **contains a value representing a specific feature** (e.g., temperature, depth, suitability score).

Raster data **can be continuous** (e.g., temperature) or **thematic** / categorical (e.g., habitat type).

SDMs rely heavily on digital raster layers to link biological occurrence data to extant environmental conditions

Raster data model

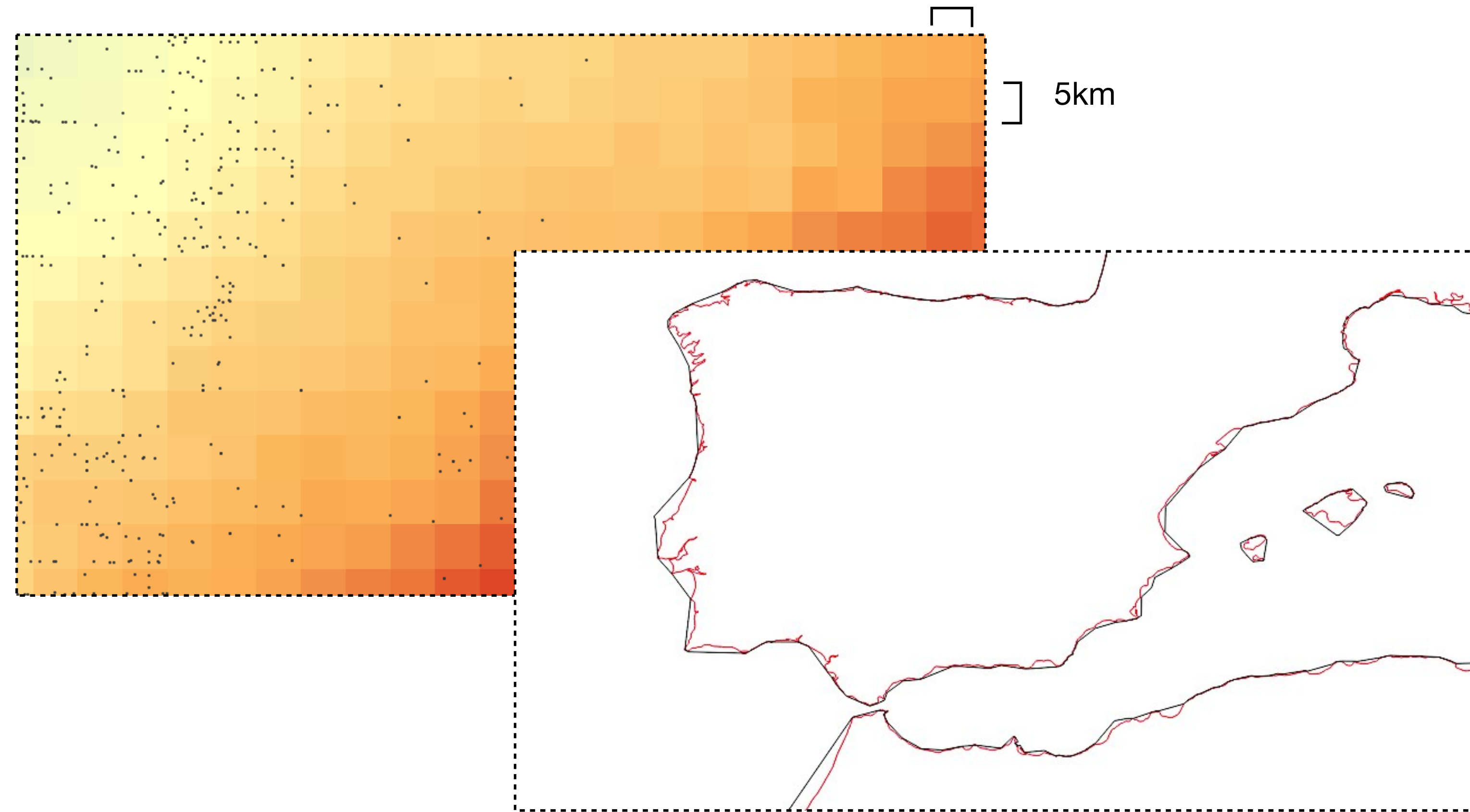
Grids and cells representing continuous features



SST in the western Mediterranean Sea and adjacent Atlantic Ocean.

Common file formats

Raster data are the the industry standard for GIS and satellite remote sensing applications. **Files are typically as .tif, .asc or .nc.**



Both models have well-defined spatial **extent and resolution**, which is the size of the **smallest feature able to be recognized**.

Vector data model: number of vertex;

Raster data model: size of cells;

Where on earth? Coordinate Reference Systems (CRS)

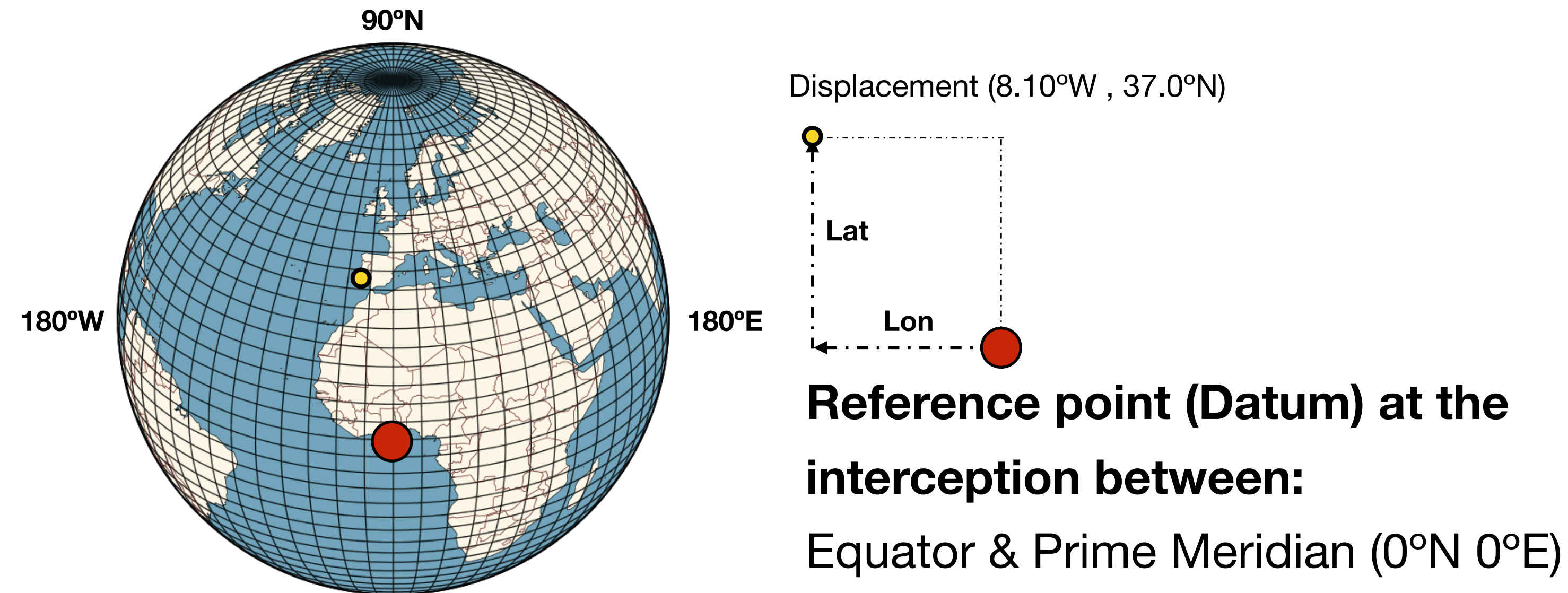
Coordinate reference systems **allow representing the locations of features within a common geographic framework.**

Without a CRS, coordinates are numbers that do not represent a specific place.

e.g., 7.321 ; -121.981

A CRS gives meaning to the information of location.

CRS have an inherent **fundamental challenge**: representing the 3D Earth on 2D surfaces.



CRS type 1: Geographic Coordinate Systems (GCS)

Defines locations directly on the 3D Earth using latitude and longitude (angles).

Components:

Datum: positions locations relative to the Earth.

Displacement units: degrees (usually decimal degrees).

Decimal degrees are simpler to use.

37°2'38.08"N ; 7°58'20.73"W

37.043911° ; -7.972425° 

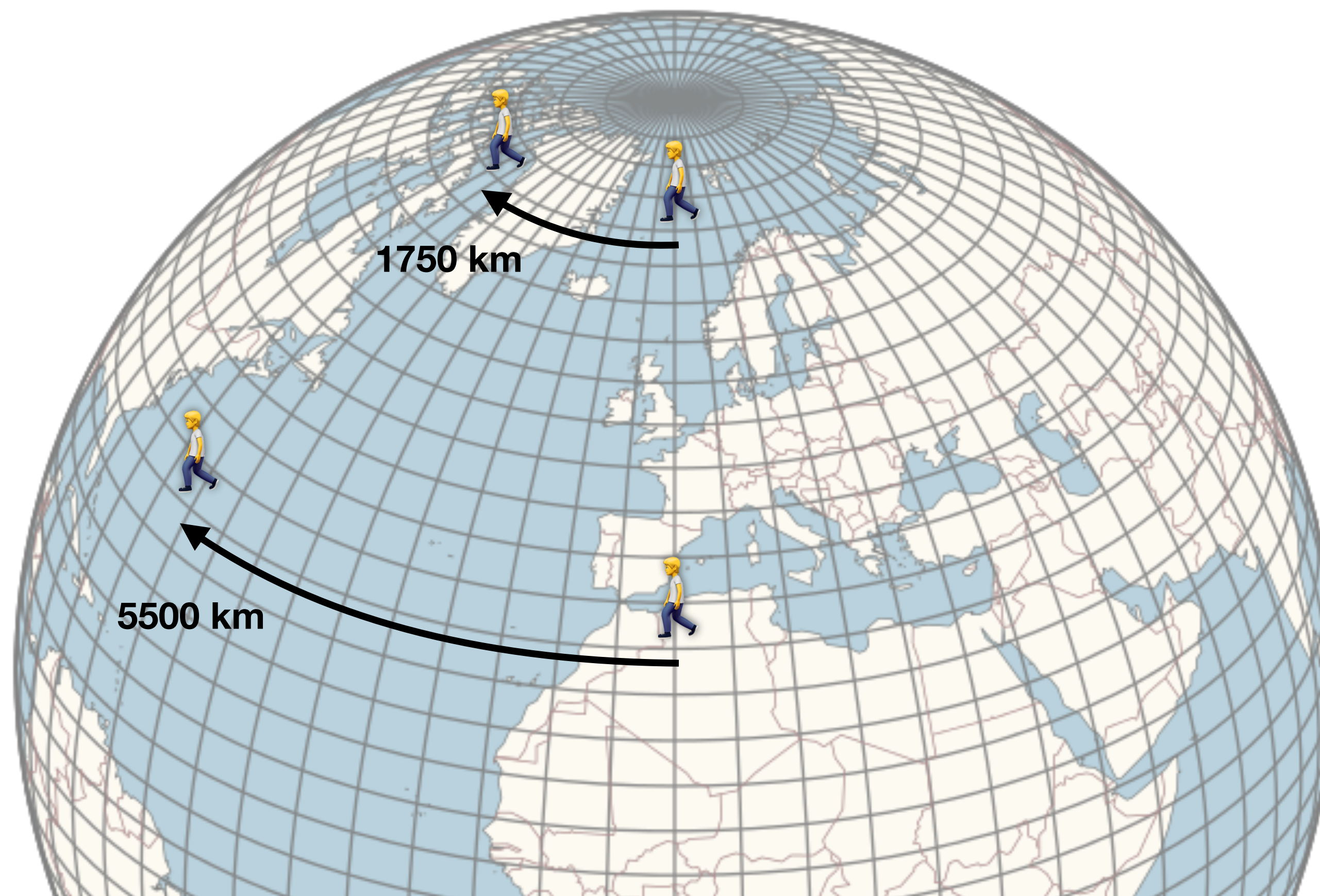
Simpler to type and share with others;

In data tables (e.g., R, Excel), longitude and latitude entries **use single cells, allowing direct analyses** (e.g., calculate distances).

Geographic coordinate systems

GCS are not good to make measurements.

There is the need to project data to an alternative coordinate system (e.g., based on actual distances).

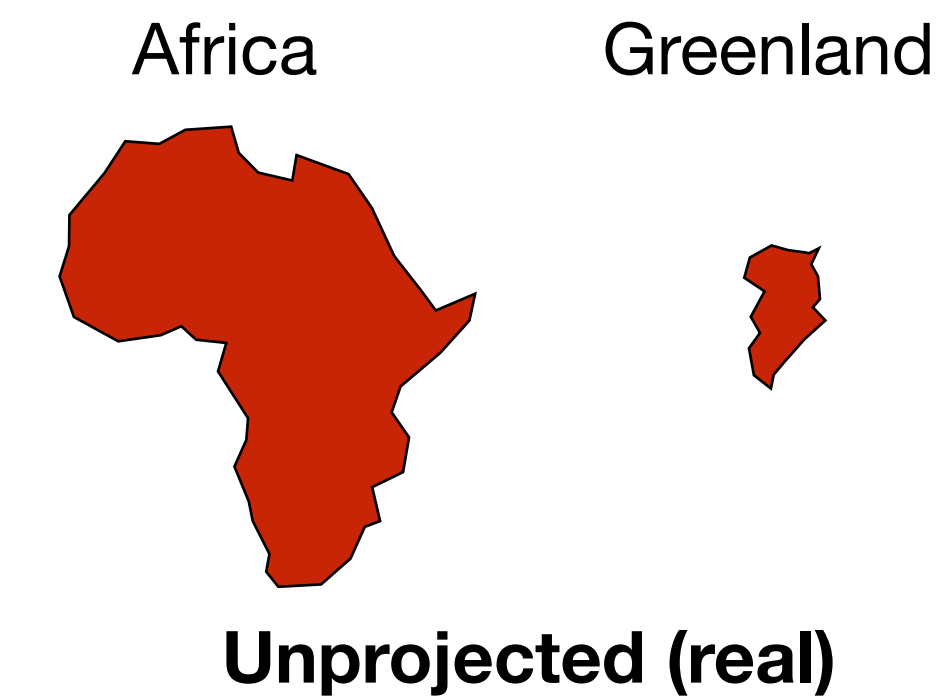
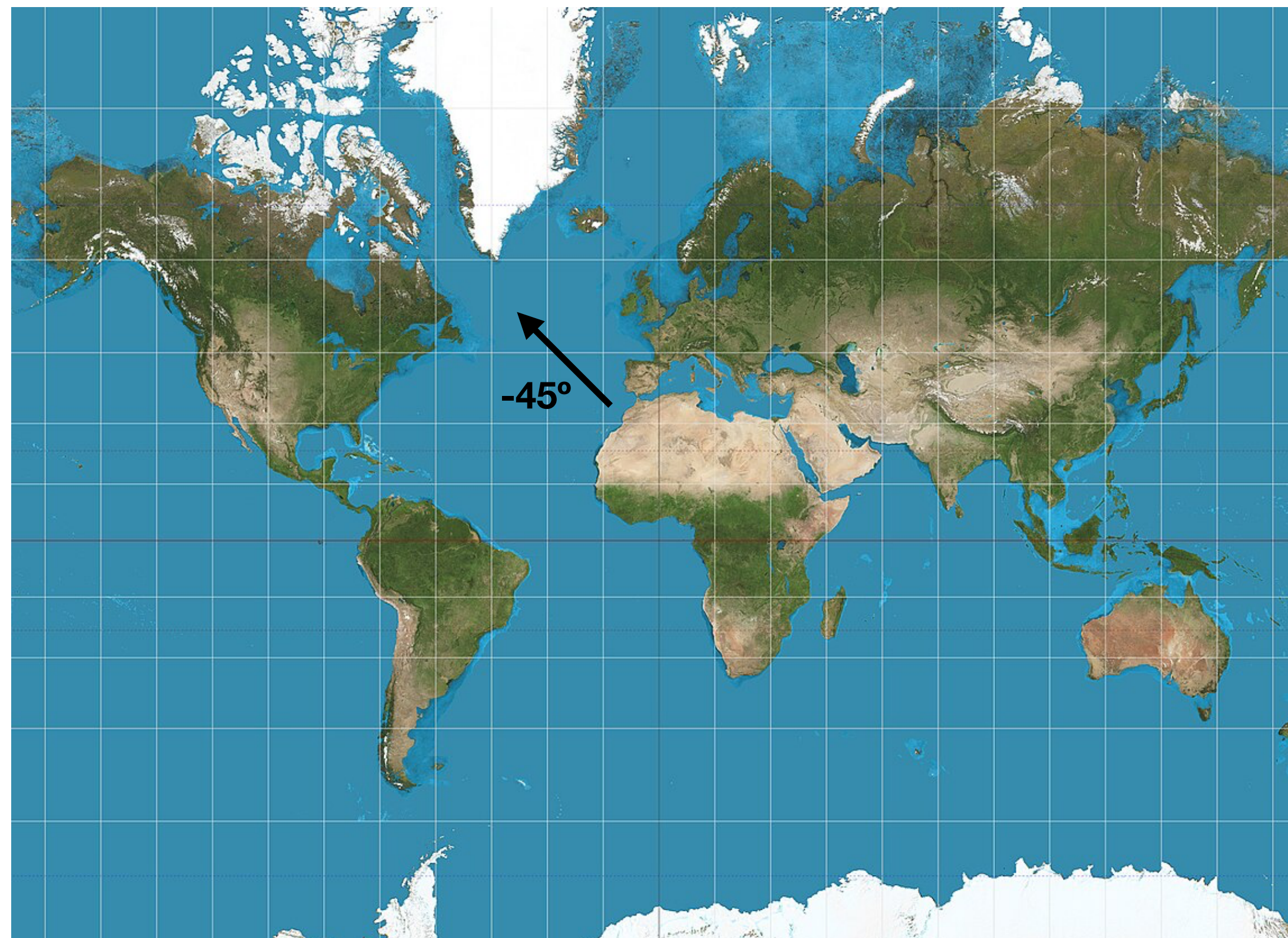


Displacement of 60° performed at different latitudes (360° longitude).

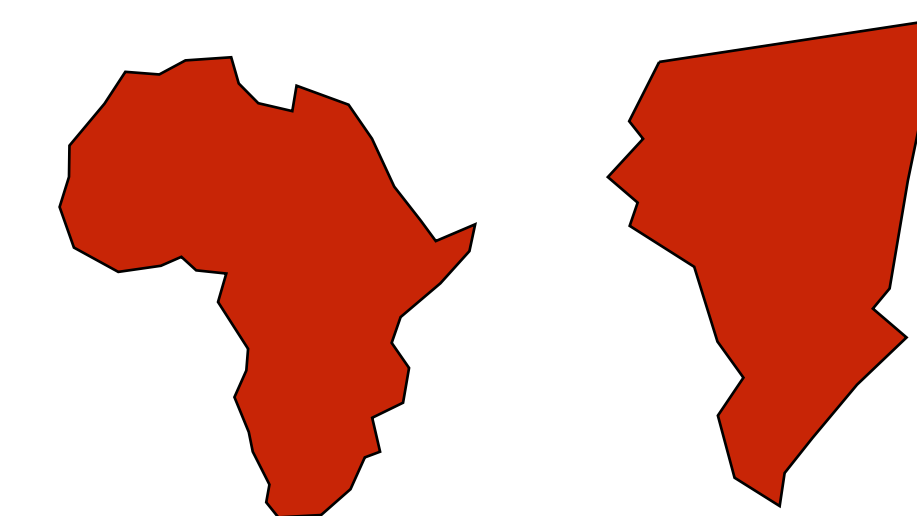
CRS type 2: Projected Coordinate Systems (PCS)

Mathematical transformation (**projection**) of the Earth's surface onto a 2-dimensional plane (**map**) with **constant units of distance (e.g., m), area, direction across the two dimensions.**

Crucially, **all map projections introduce distortion.** It's impossible to perfectly flatten a curved surface. Projections inevitably distort one or more properties.

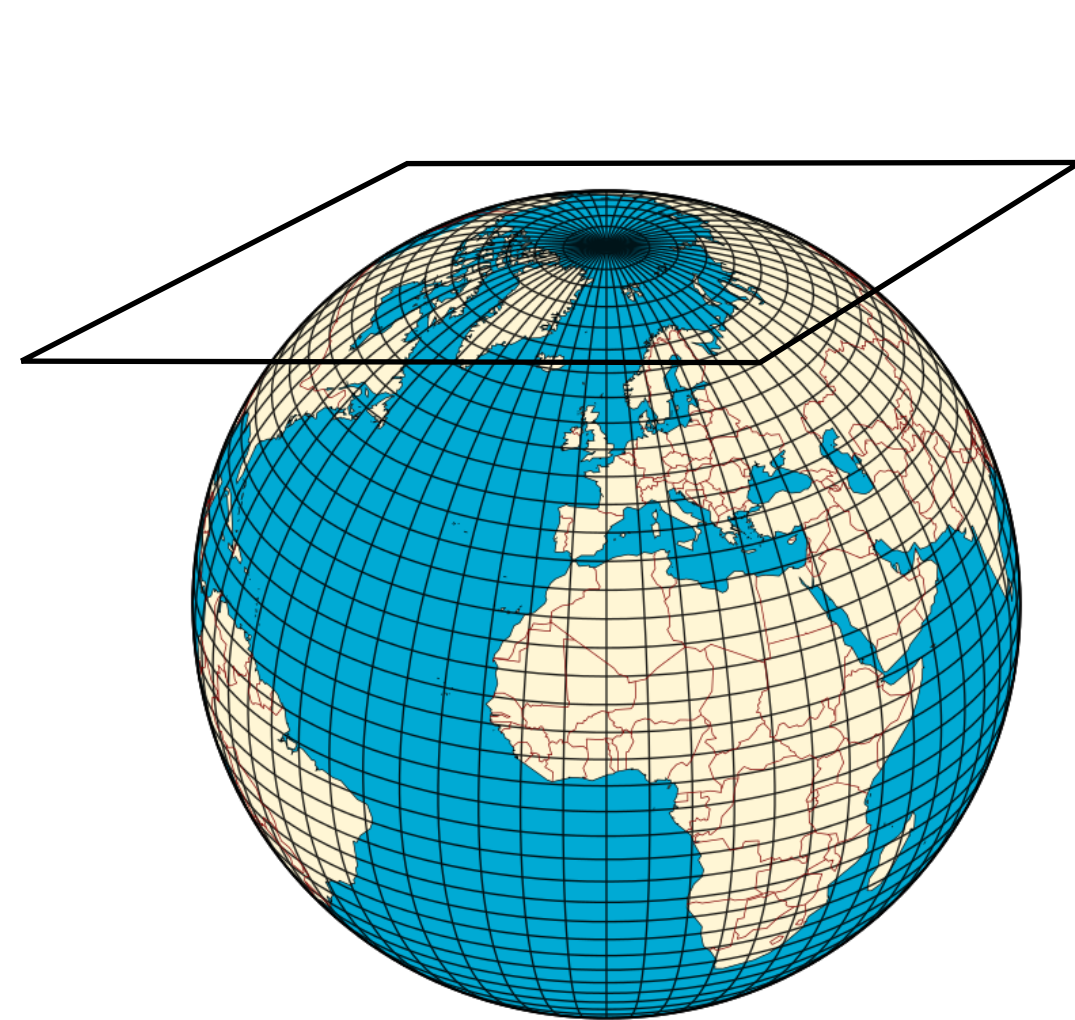


Mercator projection

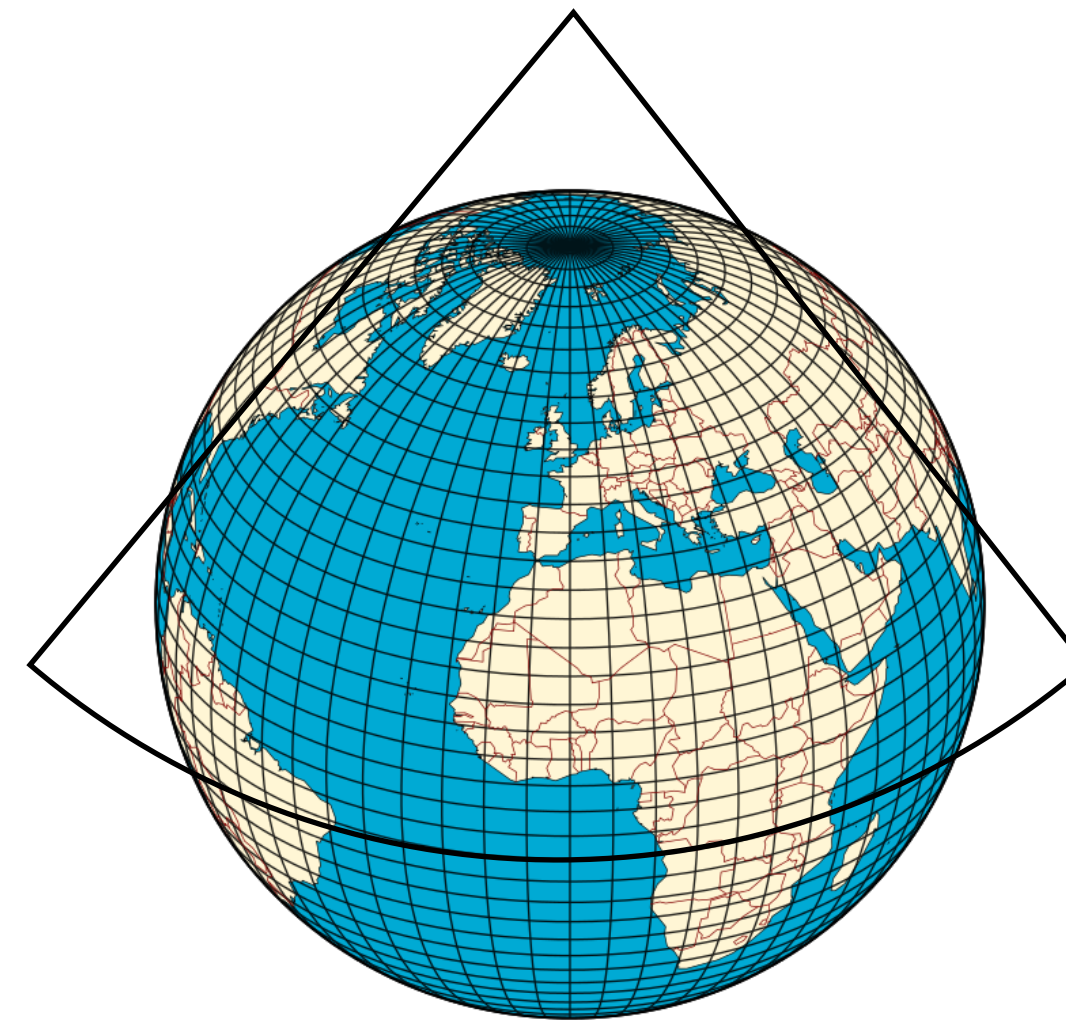


Mercator is a cylindrical projected coordinate systems that **represents lines of constant angle** (i.e., course; good for nautical use).

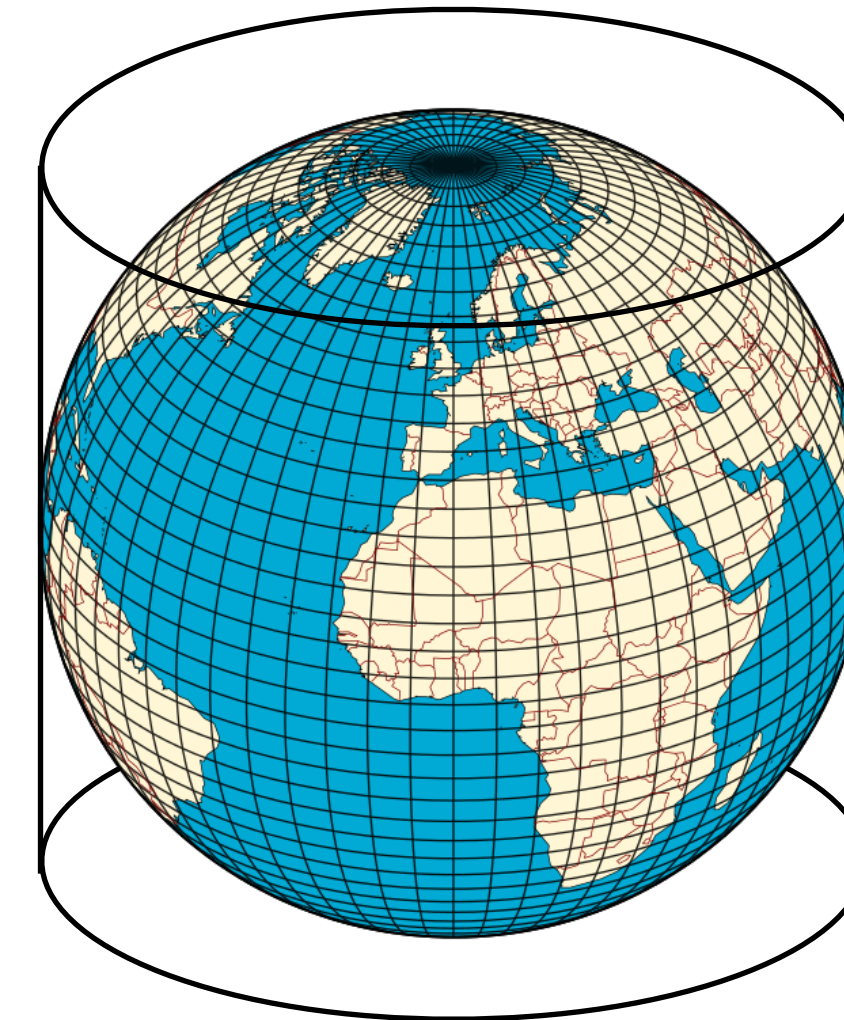
Mercator projection, distortion increases towards higher latitudes.



**Planar projection
(poles)**



**Conical projection
(mid-latitudes)**



**Cylindrical projection
(rectangular maps)**

There is no best projection. **Map-maker must decide which property is important** for the application (e.g., equal area for area calculations, equidistant for distance measurements).

What can GIS do in a SDM context?

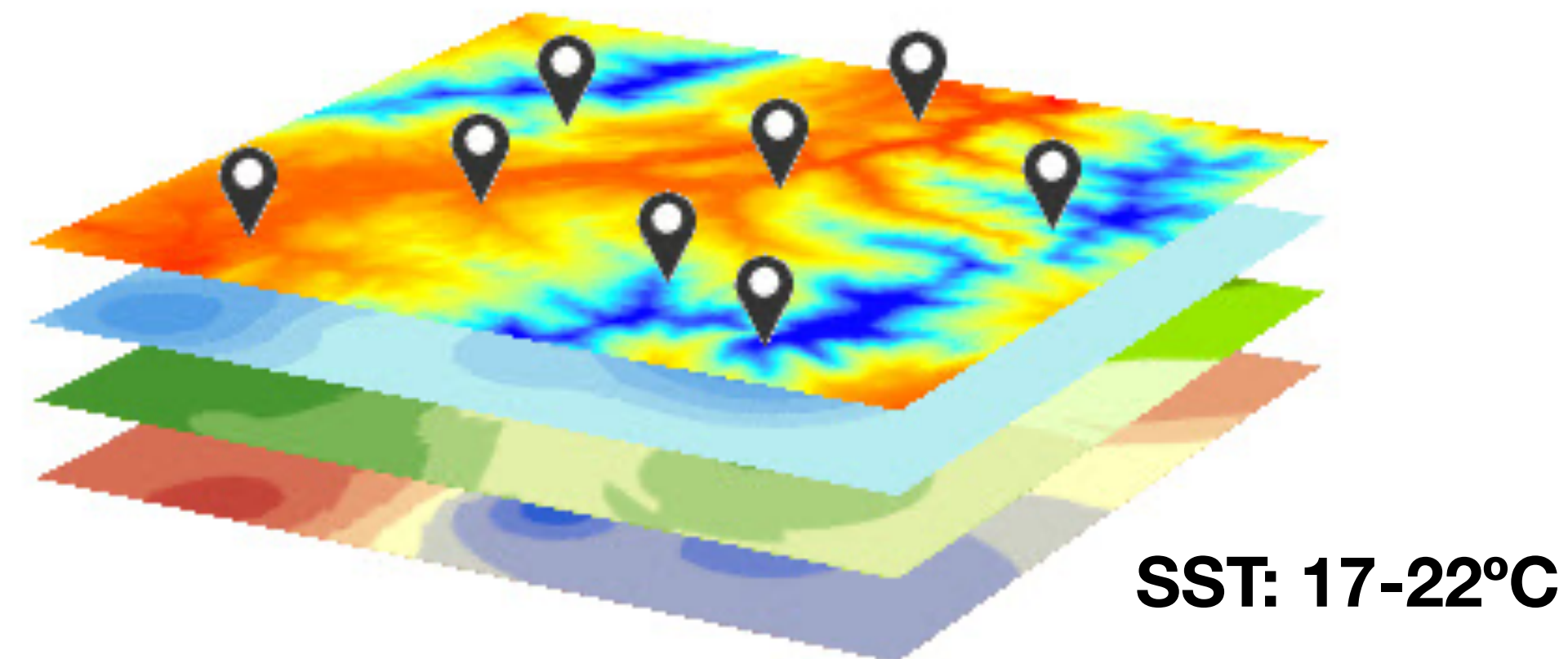
Data Preparation & Cleaning: e.g., species occurrence records (vectors) mistakenly recorded out of their known distribution.

Environmental Extraction: Using the geographic coordinates of species occurrences to **extract the exact environmental values** (e.g., temperature, precipitation) from underlying continuous raster layers to build the modeling datasets.

Spatial Cropping & Masking: Trimming global environmental raster layers **to the specific study area**, focusing the model on accessible habitats.

Mapping & Visualization: Converting complex, continuous mathematical model outputs into **visual maps** - such as habitat suitability heatmaps or binary presence/absence maps - that can inform conservation and policy.

Biological Data for SDMs



Occurrence records for modeling

Occurrence records allow building SDMs and answer *why* a species occurs in a given area.

Essential components: Must reference a particular taxon, at a particular place (geographic coordinates), and (ideally) at a particular time.

Metadata can describe the source of record (sampling method), means of identification (e.g., visual census), and spatial precision of record (maximum error).



Occurrence data for SDM: one species example

Data

Metadata

Lon	Lat	Date	Depth	Species	Common name	Behaviour	Observer
0,61287	39,58955	10/08/12	22	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
0,83999	39,77417	03/03/12	12	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
3,18192	42,40479	14/08/11	1	Caretta caretta	Loggerhead sea turtle	Eating	Unknown
-4,86395	55,24295	01/01/04	0	Caretta caretta	Loggerhead sea turtle	Swimming	Unknown
-5,89297	57,06977	03/03/12	0	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
34,9082	32,55822	22/12/02	23	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
0,61287	39,58955	14/08/11	12	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
0,83999	39,77417	03/03/12	2	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
3,18192	42,40479	22/12/02	5	Caretta caretta	Loggerhead sea turtle	Eating	Unknown
-4,86395	55,24295	03/03/12	7	Caretta caretta	Loggerhead sea turtle	Swimming	Unknown
-5,89297	57,06977	01/01/04	3	Caretta caretta	Loggerhead sea turtle	Eating	Unknown

Where to Source Biological Data

Global Biodiversity information centres;

Peer-reviewed literature;

Reference collections (e.g. museums and herbaria);

Citizen science initiatives.

Occurrence records: Global Biodiversity information centers

(i.e., databases linked via a common access portal).

Global Biodiversity Information Facility (GBIF; <https://www.gbif.org/>):

Occurrence records: over 2.7 billion

Ocean Biodiversity Information System (OBIS; <https://obis.org/>):

Occurrence records: over 130 million

National Biodiversity Network (NBN; UK; <https://nbnatlas.org/>):

Occurrence records: over 320 million

Atlas of Living Australia (ALA; <https://www.ala.org.au/>):

Occurrence records: over 130 million

Occurrence records: Literature

Literature is useful to compile biodiversity records.

Peer-reviewed publications are preferred, even when out of the scope of our work.

The screenshot shows the Web of Science search results page. The top navigation bar includes links for Web of Science™, InCites®, Journal Citation Reports®, Essential Science Indicators™, and EndNote®, along with Sign In, Help, and English options. The main header displays the Web of Science™ logo and the Thomson Reuters™ logo. Below the header, there is a 'Back to Search' button and links for My Tools, Search History, and Marked List. The search results are displayed in a list format, sorted by Publication Date -- newest to oldest. The results show 1,766 results from all databases. The search criteria are: TOPIC: ("Caretta caretta"). The results list includes three entries, each with a title, author information, publication details, and a 'View Abstract' button. The first entry is 'Stable isotopic comparison between loggerhead sea turtle tissues' by Vander Zanden, Hannah B; Tucker, Anton D; Bolten, Alan B; et al., published in Rapid communications in mass spectrometry : RCM, Volume 28, Issue 19, Pages 2059-64, Published: 2014-Oct-15. The second entry is 'The geomagnetic environment in which sea turtle eggs incubate affects subsequent magnetic navigation behaviour of hatchlings' by Fuxjager, Matthew J; Davidoff, Kyla R; Mangiamele, Lisa A; et al., published in Proceedings. Biological sciences / The Royal Society, Volume 281, Issue 1791, Published: 2014-Sep-22. The third entry is 'Levels and profiles of POPs (organochlorine pesticides, PCBs, and PAHs) in free-ranging common bottlenose dolphins of the Canary Islands, Spain' by Garcia-Alvarez, Natalia; Martin, Vidal; Fernandez, Antonio; et al., published in SCIENCE OF THE TOTAL ENVIRONMENT, Volume 493, Pages 22-31, Published: SEP 15 2014. The left sidebar contains a 'Refine Results' section with a search box and a 'Databases' section. The 'Research Domains' section includes checkboxes for SCIENCE TECHNOLOGY, SOCIAL SCIENCES, and ARTS HUMANITIES.

Web of Science™ InCites® Journal Citation Reports® Essential Science Indicators™ EndNote® Sign In Help English

WEB OF SCIENCE™ THOMSON REUTERS™

Back to Search My Tools Search History Marked List

Results: 1,766
(from All Databases)

You searched for: TOPIC: ("Caretta caretta") ...More

Refine Results

Search within results for...

Databases

Research Domains

☐ SCIENCE TECHNOLOGY
☐ SOCIAL SCIENCES
☐ ARTS HUMANITIES

Sort by: Publication Date -- newest to oldest

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☐ Select Page

1. **Stable isotopic comparison between loggerhead sea turtle tissues.**
By: Vander Zanden, Hannah B; Tucker, Anton D; Bolten, Alan B; et al.
Rapid communications in mass spectrometry : RCM Volume: 28 Issue: 19 Pages: 2059-64 Published: 2014-Oct-15
 Times Cited: 0 (from All Databases)

2. **The geomagnetic environment in which sea turtle eggs incubate affects subsequent magnetic navigation behaviour of hatchlings.**
By: Fuxjager, Matthew J; Davidoff, Kyla R; Mangiamele, Lisa A; et al.
Proceedings. Biological sciences / The Royal Society Volume: 281 Issue: 1791 Published: 2014-Sep-22
 Times Cited: 0 (from All Databases)

3. **Levels and profiles of POPs (organochlorine pesticides, PCBs, and PAHs) in free-ranging common bottlenose dolphins of the Canary Islands, Spain**
By: Garcia-Alvarez, Natalia; Martin, Vidal; Fernandez, Antonio; et al.
SCIENCE OF THE TOTAL ENVIRONMENT Volume: 493 Pages: 22-31 Published: SEP 15 2014
 Times Cited: 0 (from All Databases)

Occurrence records: Literature

Literature is useful to compile biodiversity records.

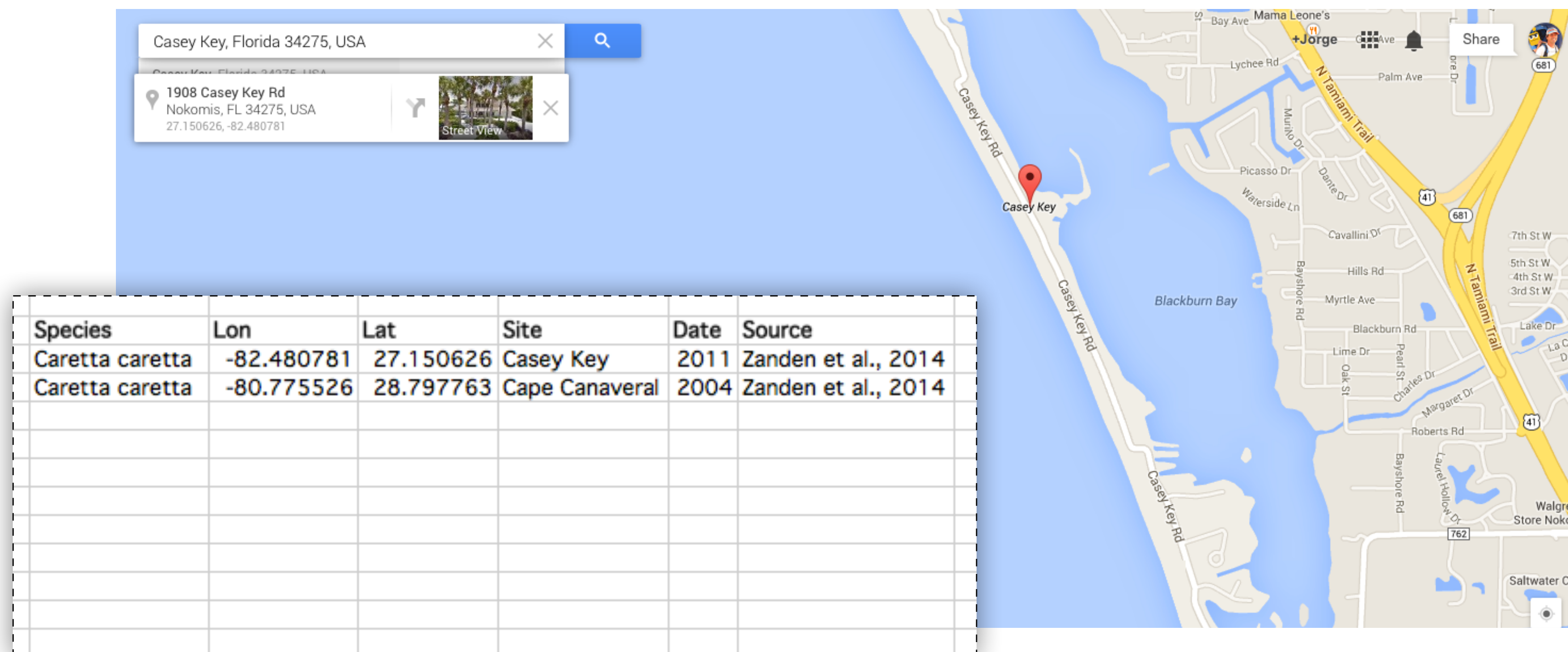
Peer-reviewed publications are preferred, even when out of the scope of our work.

The screenshot shows the Web of Science interface. At the top, there's a navigation bar with links to Web of Science™, InCites®, Journal Citation Reports®, Essential Science Indicators™, and EndNote®, along with Sign In, Help, and English options. The main header displays 'WEB OF SCIENCE™' and the Thomson Reuters logo. Below this, a 'Back to Search' button is visible. The search results section indicates 'Results: 1,766 (from All Databases)' and shows the search criteria: 'You searched for: TOPIC: ("Caretta caretta") ...More'. A 'Refine Results' section on the left includes a search bar and filters for Databases and Research Domains (Science Technology, Social Sciences, Arts Humanities). The main results list shows three entries. The first entry, 'Field collections', is highlighted with a dashed box and contains the text: 'Nesting female loggerheads were sampled at Cape Canaveral National Seashore (CNS), on the east coast of Florida in 2004 (n = 13), and Casey Key (CK), on the west coast of Florida, in 2011 (n = 20) between the months of May and July. Prior to collection, the sampling site was cleaned with isopropyl alcohol. Scute and skin were collected with 5 or 6 mm biopsy punches. The skin samples contained both epidermal and...'. The second entry is 'The geom... navigation' by Fuxjage, and the third is 'Levels and profiles of POPs (organochlorine pesticides, PCBs, and PAHs) in free-ranging common bottlenose dolphins of the Canary Islands, Spain' by Garcia-Alvarez, Natalia; Martin, Vidal; Fernandez, Antonio; et al. The third entry also shows citation information: 'SCIENCE OF THE TOTAL ENVIRONMENT Volume: 493 Pages: 22-31 Published: SEP 15 2014' and 'Times Cited: 0 (from All Databases)'.

Occurrence records: Literature

Literature is useful to compile biodiversity records.

Peer-reviewed publications are preferred, even when out of the scope of our work.



Occurrence records: Biological reference collections

Preserved specimens / photographs maintained in **museums, universities, botanic gardens**, and similar **institutions**.

A major tools for basic investigation and assessment of biodiversity.

~2.5 billion specimens in preserved collections.

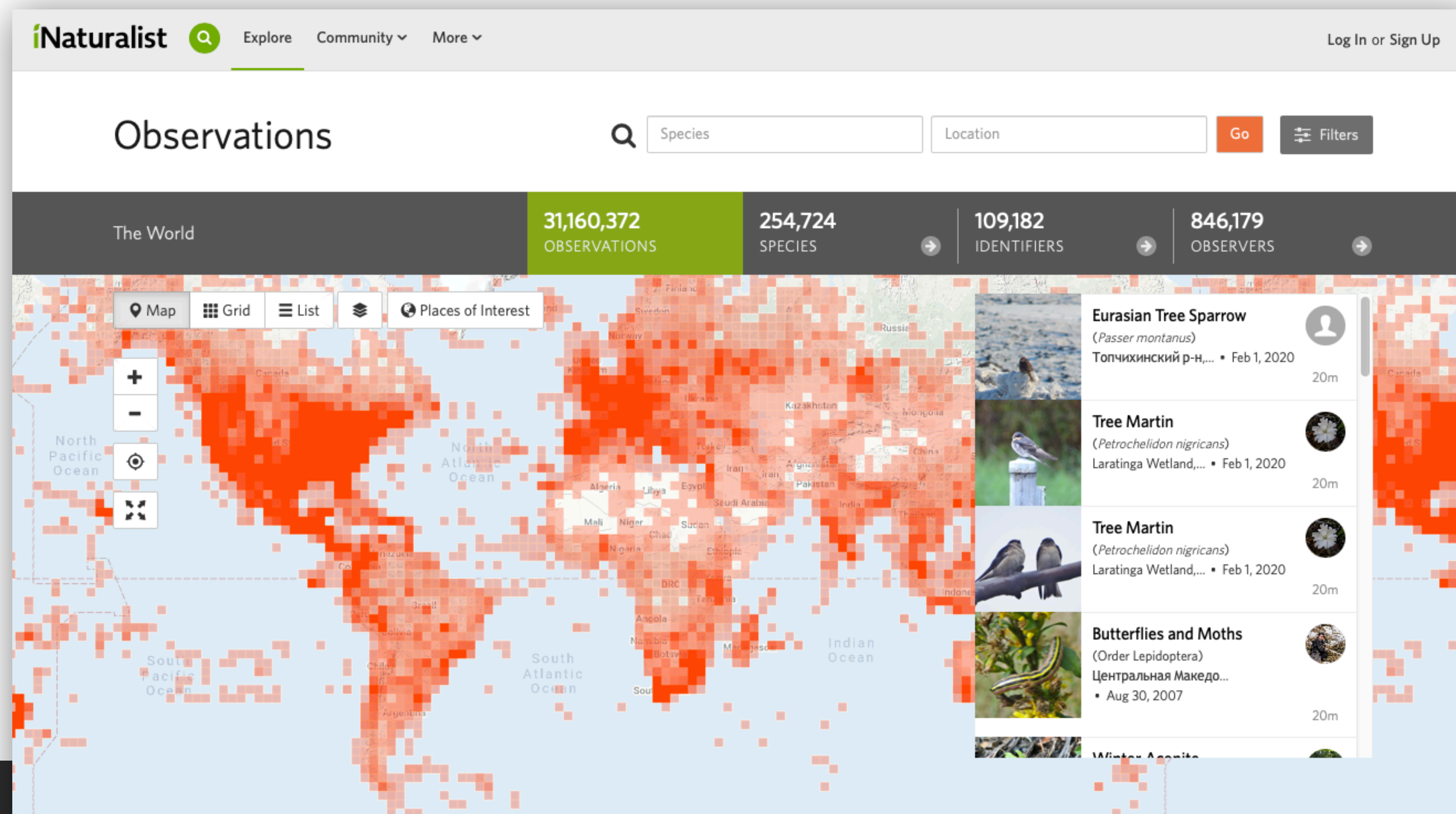
Macroalgae Herbarium Portal (www.macroalgae.org):

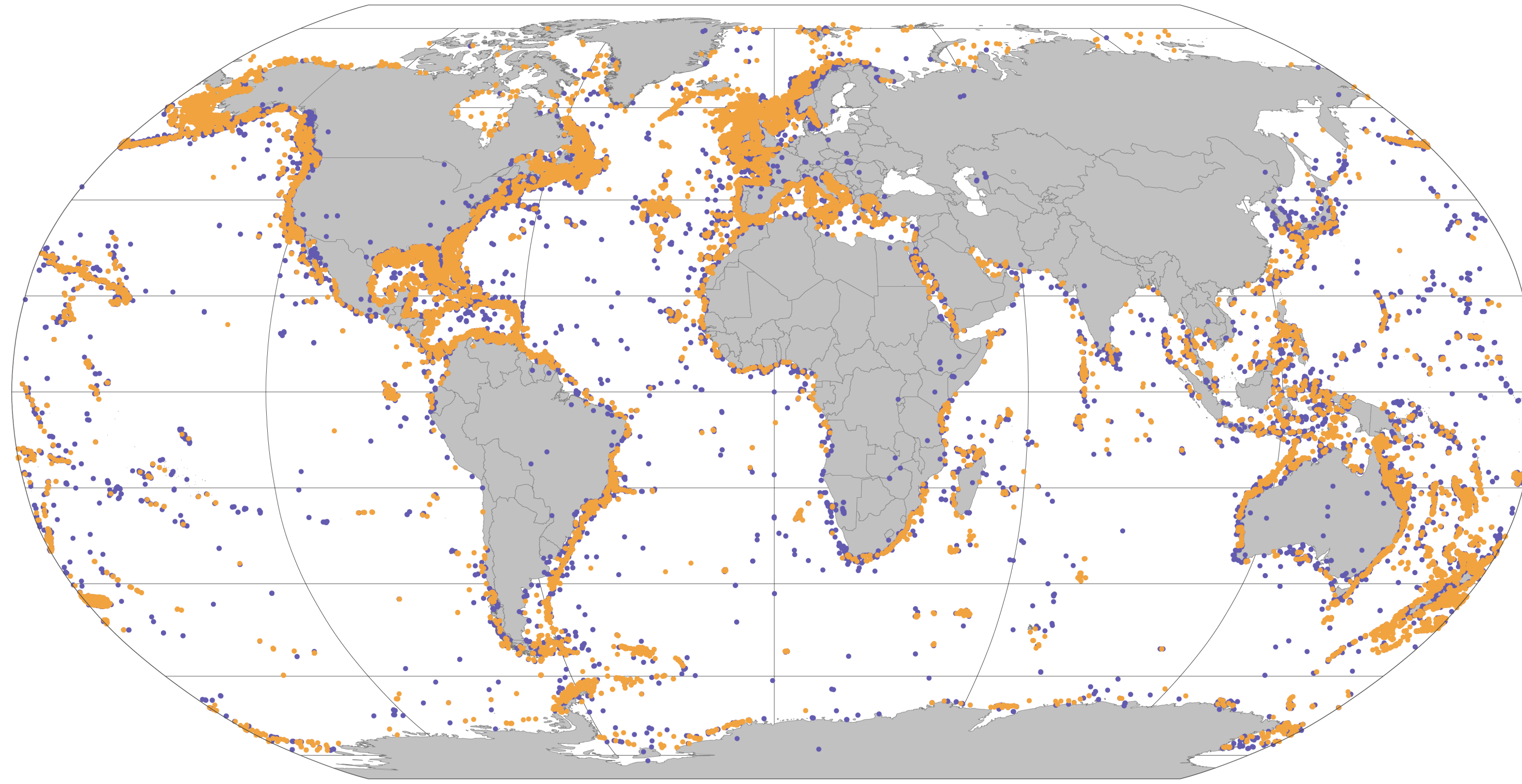
Occurrence records: >1 million records.

Occurrence records: Citizens can help

Sharing biodiversity records / pictures with scientists.

iNaturalist (<http://www.inaturalist.com>): > 300 million records.



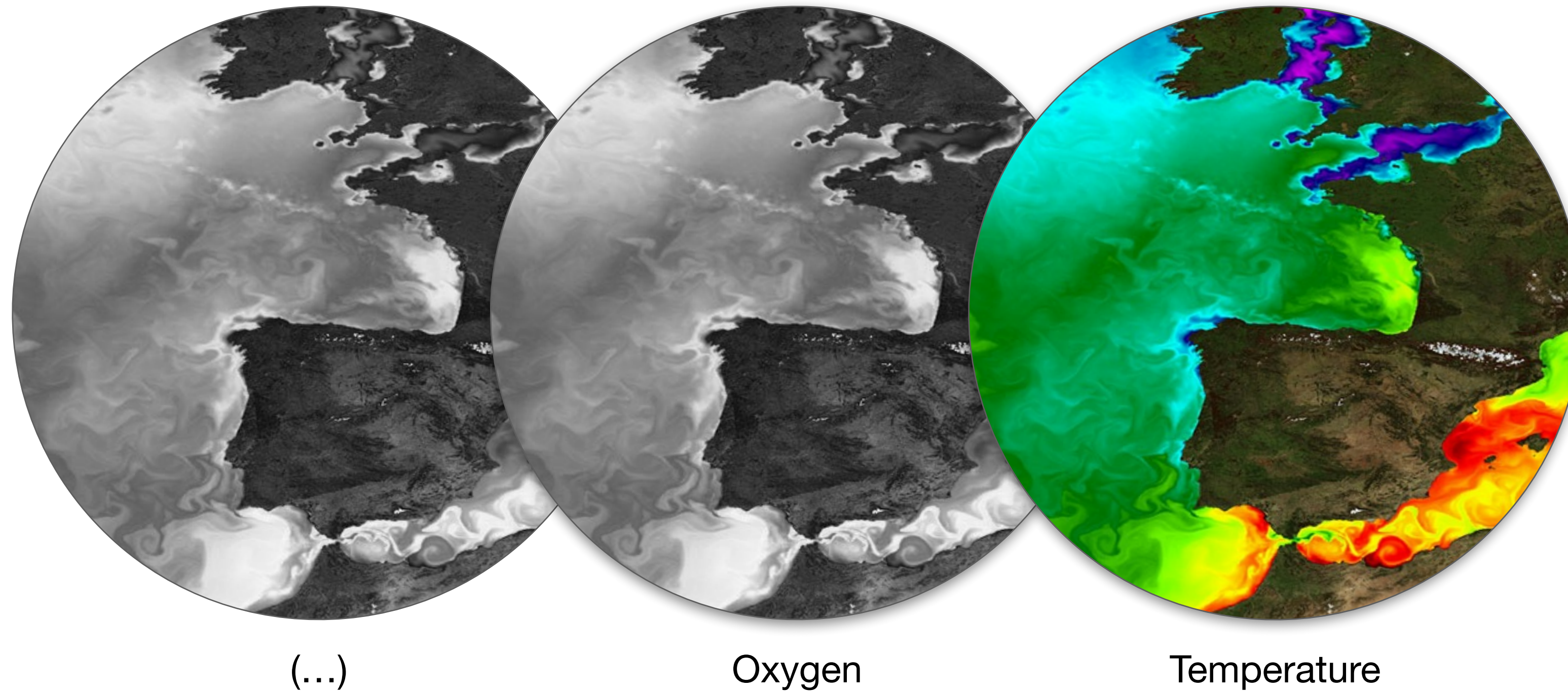


4 million records of ~1,500 species of kelp, seagrass and cold-water corals.

Data gathered from **literature** and major **biodiversity information facilities**.

Quality control of records based on published and expert knowledge to **eliminate entries outside the known vertical and geographic distribution of species.**

Environmental Layers for SDMs



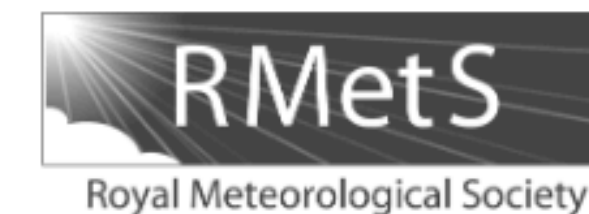
Environmental data

Allows **linking biodiversity records with the conditions they were exposed to.**

Environmental predictors typically in the form of digital raster layers.

For the **terrestrial realm**, GIS databases like **World-Clim** provide uniform resolution and global coverage. Incorporates CMIP6 Earth System Models and Shared Socioeconomic Pathways (SSPs), making it the standard for terrestrial SDMs.

INTERNATIONAL JOURNAL OF CLIMATOLOGY
Int. J. Climatol. **37**: 4302–4315 (2017)
 Published online 15 May 2017 in Wiley Online Library
 (wileyonlinelibrary.com) DOI: 10.1002/joc.5086



WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas

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ABSTRACT: We created a new dataset of spatially interpolated monthly climate data for global land areas at a very high spatial resolution (approximately 1 km²). We included monthly temperature (minimum, maximum and average), precipitation, solar radiation, vapour pressure and wind speed, aggregated across a target temporal range of 1970–2000, using data from between 9000 and 60 000 weather stations. Weather station data were interpolated using thin-plate splines with covariates including elevation, distance to the coast and three satellite-derived covariates: maximum and minimum land surface temperature as well as cloud cover, obtained with the MODIS satellite platform. Interpolation was done for 23 regions of varying size depending on station density. Satellite data improved prediction accuracy for temperature variables 5–15% (0.07–0.17 °C), particularly for areas with a low station density, although prediction error remained high in such regions for all climate variables. Contributions of satellite covariates were mostly negligible for the other variables, although their importance varied by region. In contrast to the common approach to use a single model formulation for the entire world, we constructed the final product by selecting the best performing model for each region and variable. Global cross-validation correlations were ≥ 0.99 for temperature and humidity, 0.86 for precipitation and 0.76 for wind speed. The fact that most of our climate surface estimates were only marginally improved by use of satellite covariates highlights the importance having a dense, high-quality network of climate station data.

KEY WORDS interpolation; climate surfaces; WorldClim; MODIS; land surface temperature; cloud cover; solar radiation; wind speed; vapour pressure

For the **marine realm**, **Bio-ORACLE provides** uniform resolution, global coverage, and biologically significant variables, incorporating also CMIP6 Earth System Models and Shared Socioeconomic Pathways (SSPs).

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DOI: 10.1111/geb.13813

DATA ARTICLE

Global Ecology and BiogeographyA Journal of MacroecologyWILEY

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected with species distribution modelling using standardized data layers representing physi-



This helps evaluate climate models, leads to improvements in the model simulations and provides a better understanding of past, present and future climates.

State-of-the-art

Layers for future conditions comprise numerous Earth System Models.

ESM are **simulations that resolve global atmospheric and oceanographic conditions** across space and time (e.g., end-of-century SST in Culatra island).

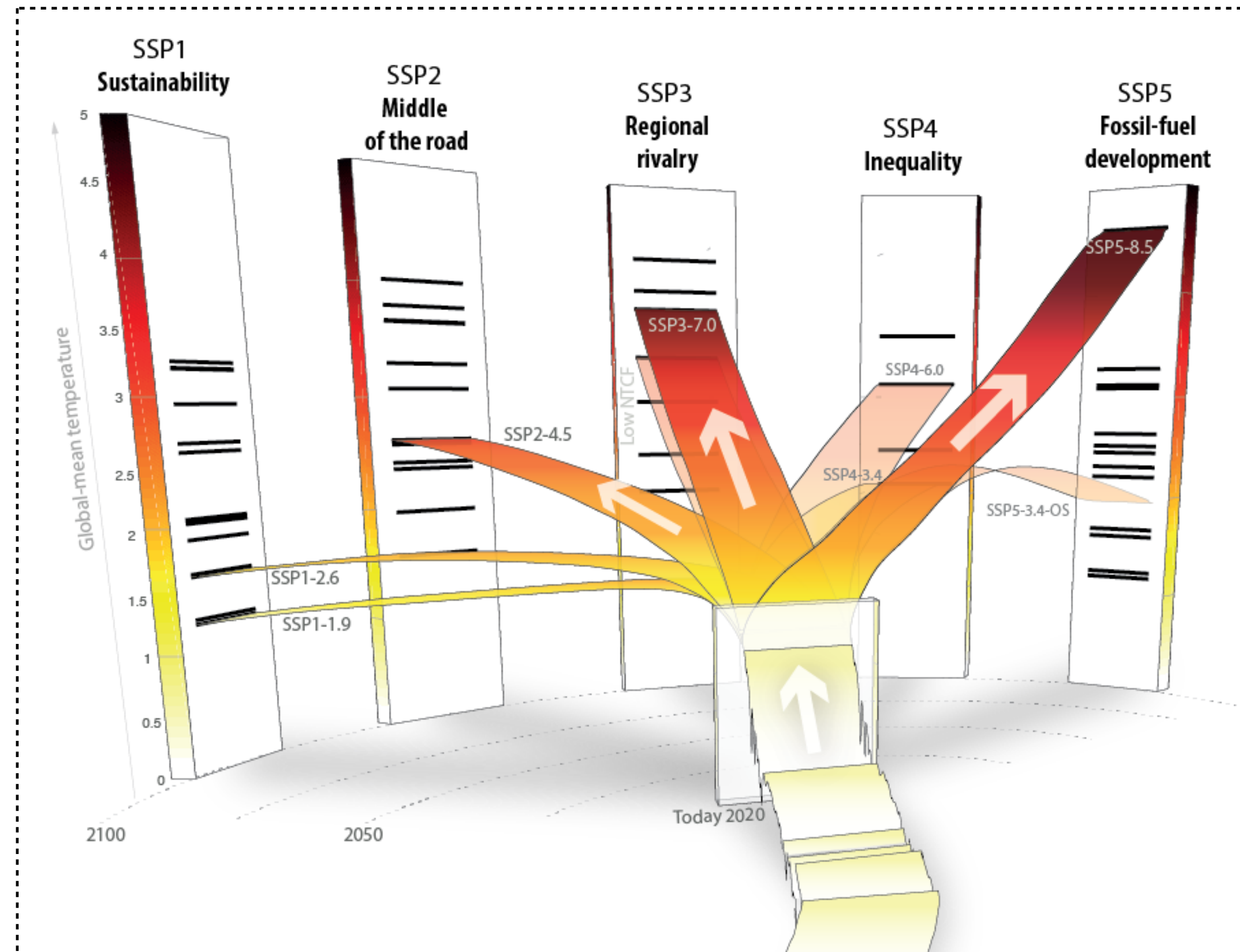
ESM **have inherent uncertainties.**

The **multi-model ensemble** (e.g., averaging different models) to **account for inter-ESM variability**. Previous available datasets limited to 2 ESMs (from the dozens available), limiting the potential to capture the range of climate change.

State-of-the-art

Layers for future conditions **consider data (ESM) from the 6th phase of the Coupled Model Inter-comparison Project.**

CMIP6 (1) **is more accurate in capturing the spatial and temporal variability of oceanographic conditions** and (2) **outputs the SSP scenarios,** which are the new narratives of global change.



The new **SSP** use up-to-date socioeconomic assumptions and encompass the **Paris Agreement** expectations, allowing the development of projections aligned with international climate policies.

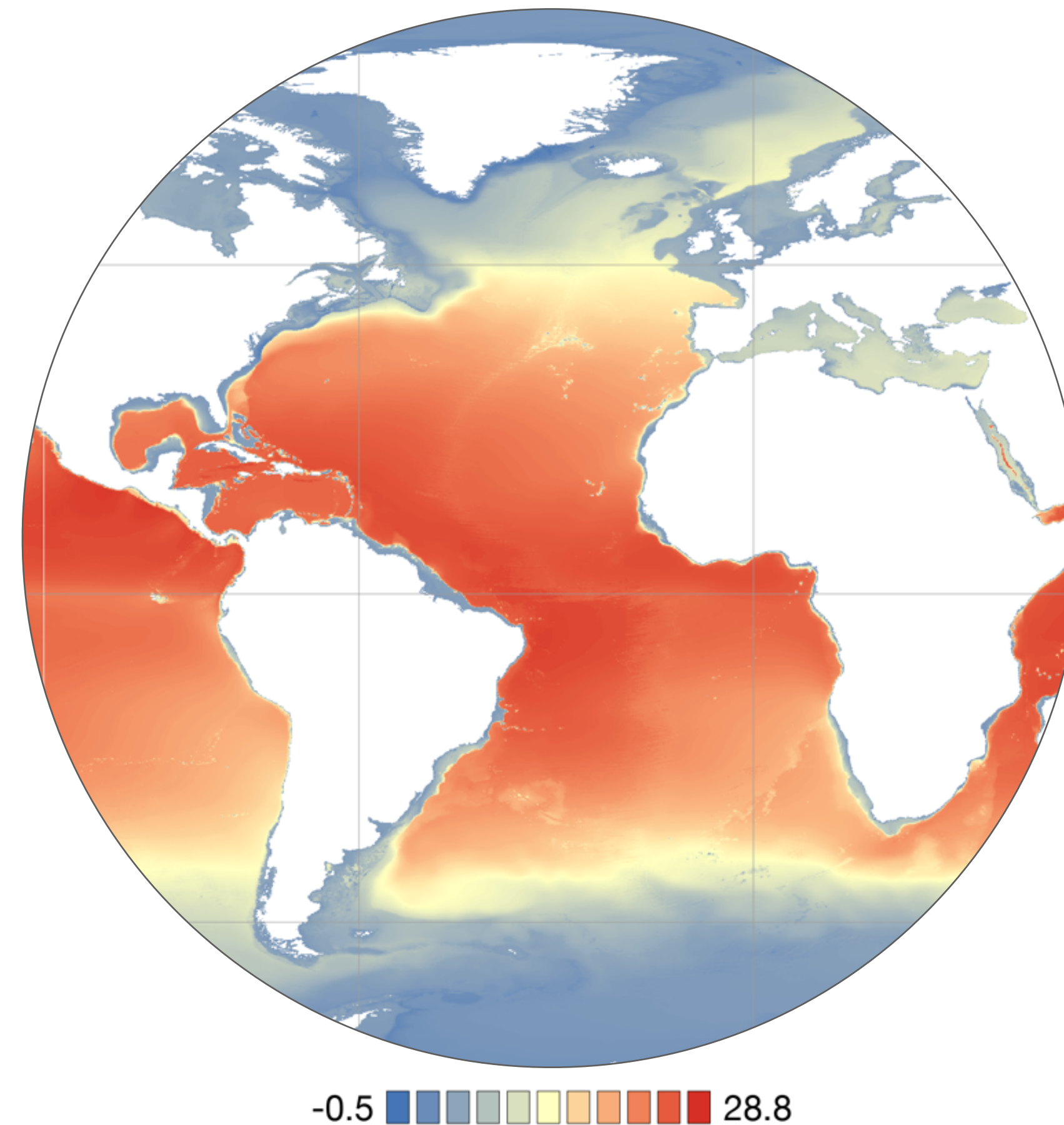
State-of-the-art

Moving beyond just temperature: While temperature is vital, relying on it exclusively misses the full physiological niche. Biologically meaningful models must incorporate variables like dissolved oxygen, primary productivity, and pH.

Layers for future conditions **consider biologically meaningful variables beyond temperature, salinity and sea ice conditions.**

Oxygen, primary productivity and pH (among others) **are limiting factors for biodiversity**, and are expected to change in the future.

Failing to consider these variables in climate change projections might **misinterpret the exposure of biodiversity to future detrimental conditions.**



-0.5 28.8
Difference between surface and bottom temperature

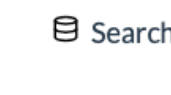
The 3D Ocean: Unlike terrestrial models, marine species are exposed to depth gradients. Sea bottom layers are key for **benthic species** (e.g., corals or fish).

		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		1 (1970-2020)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)
1	Air temperature							
2	Precipitation							

WorldClim 2.1 (CIMP6): 2 terrestrial bioclimatic variables from 1970-2100, by 20-year averages at 0.5 arc-minutes resolution (~1km), covering the span of SSPs, by ensembling 23 GCMs of CMIP6.

		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		2 (2000-2020)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)
1	Temperature							
2	Salinity							
3	Sea Ice Cover							
4	Sea Ice Thickness							
5	Sea Water Velocity							
6	Mixed Layer Depth							
7	Diffuse Attenuation Coefficient							
8	PAR							
9	PAR at Bottom							
10	Oxygen							
11	pH							
12	Iron							
13	Phosphate							
14	Nitrate							
15	Silicate							
16	Primary Production							
17	Phytoplankton							
18	Chlorophyll							

Bio-ORACLE provides **18 essential variables** provided for the surface and along the seafloor, **from 2000 to 2100, per decade**, at **0.05° resolution (~5km)**, **covering the span of SSPs**, by **ensembling 10 ESMs of CMIP6**.


BETA



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All Area

From

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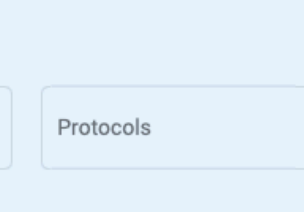
☐ Only the whole selected time range
 ☒ Only with depth level

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Global Ocean 1/12° Physics Analysis And Forecast Updated Daily

GLOBAL_ANALYSIS_FORECAST_PHY_001_024

T bottomT S SSH UV UV MLD SIC SIT SIUV

From

2019-01-01

To

Present



Global Ocean Biogeochemistry Analysis And Forecast

GLOBAL_ANALYSIS_FORECAST_BIO_001_028

CHL PHYC O2 NO3 PO4 SI FE SPCO2 PH PP

From

2019-05-04

To

Present



Global Ocean Physics Reanalysis

GLOBAL_MULTIYEAR_PHY_001_030

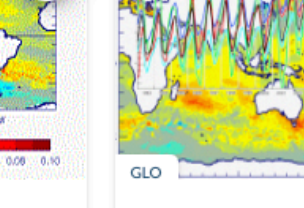
T bottomT S SSH UV MLD SIC SIT SIUV

From

1993-01-01

To

2020-05-31



Global Ocean Ensemble Physics Reanalysis - Low Resolution

GLOBAL_REANALYSIS_PHY_001_026

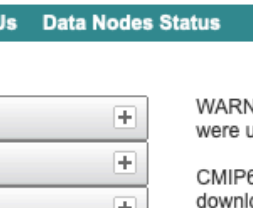
T S UV SIC SIT

From

1993-01-01

To

2019-12-15



WCRP

CMIP6

World Climate Research Programme

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Data Nodes Status

You are at the [ESGF@DOE/LLNL node](#)

Technical Support

MIP Era

Activity

Product

Source ID

Institution ID

Source Type

Nominal Resolution

Experiment ID

Sub-Experiment

Variant Label

Grid Label

Table ID

Frequency

Realm

Variable

CF Standard Name

Data Node

WARNING: Not all models include a variant "r1i1p1f1", and across models, identical values of variant_label do not imply identical variants! To learn which forcing datasets were used in each variant, please check modeling group publications and documentation provided through ES-DOC.

CMIP6 project data downloads are unrestricted. Downloads should be performed with the -s option to a wget script without the need to login. When using this method for download, ensure you are not using additional options, eg. -s and -H should never be combined.

For more information about CMIP6 data please consult this guide: <https://pcmdi.llnl.gov/CMIP6/Guide/dataUsers.html>

Please try our updated ESGF web application (named "Metagrid"), now undergoing beta testing. For this test release we are reaching out for help from users in the community to report any issues they encounter with the application. The beta-test web application can be found at the following site: <https://aims2.llnl.gov/metagrid/>
Please see the following page for more information including a FAQ: <https://esgf.github.io/esgf-user-support/metagrid.html>

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1. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.Amon.wap.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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2. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.clt.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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3. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hfls.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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4. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hfss.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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5. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hursmax.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

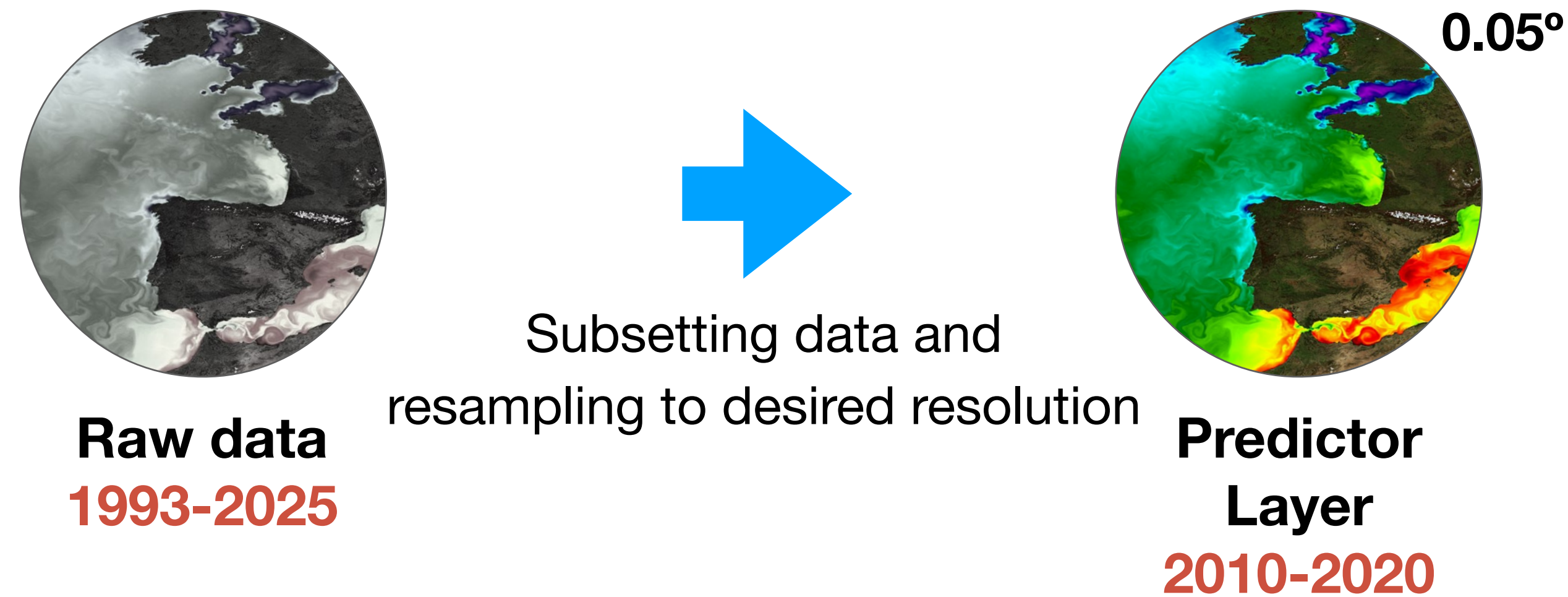
Full Dataset Services: [Show Metadata](#) [List Files](#) [WGET Script](#) [LAS](#) [Show Citation](#) [PID](#) [Further Info](#)

6. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hursmin.gn

Data Node: crd-esgf-drc.ec.gc.ca

Examples of Data Sources:

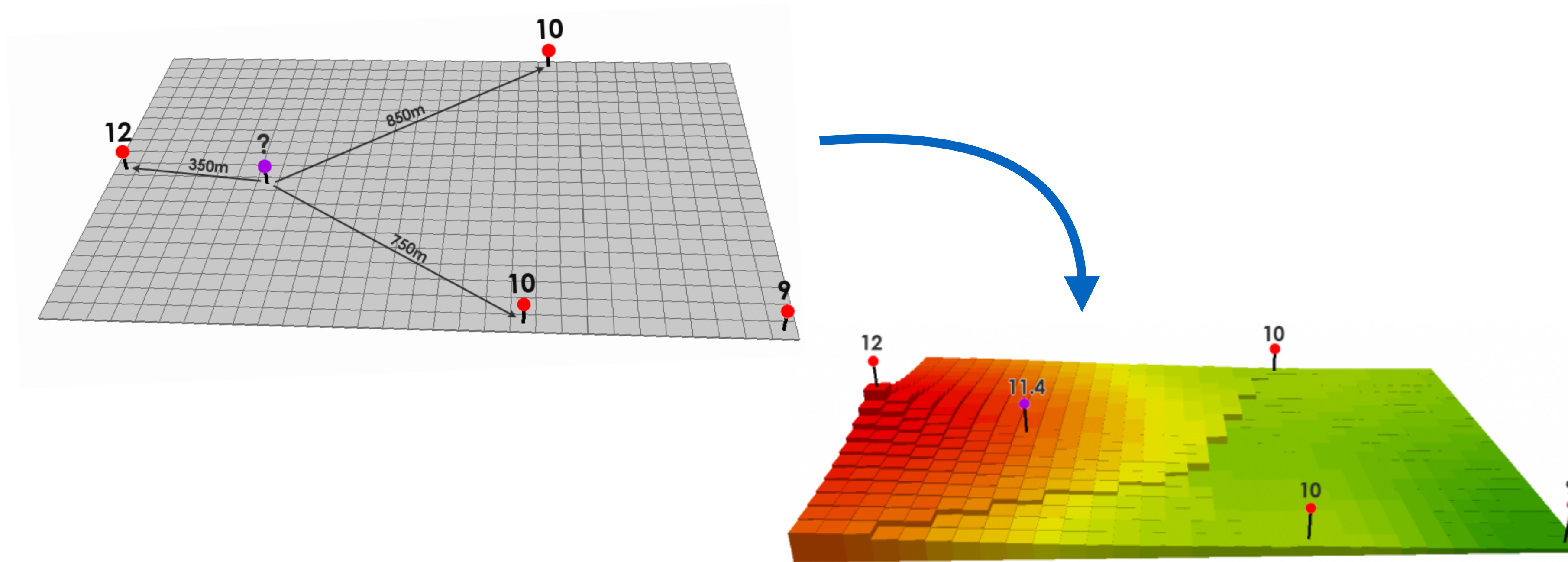
Copernicus Marine Service and World Climate Research Programme



For present-day conditions, layers are developed by subsetting raw data to the desired decade and resampling (interpolating) to the desired resolution (0.05°).

e.g.,

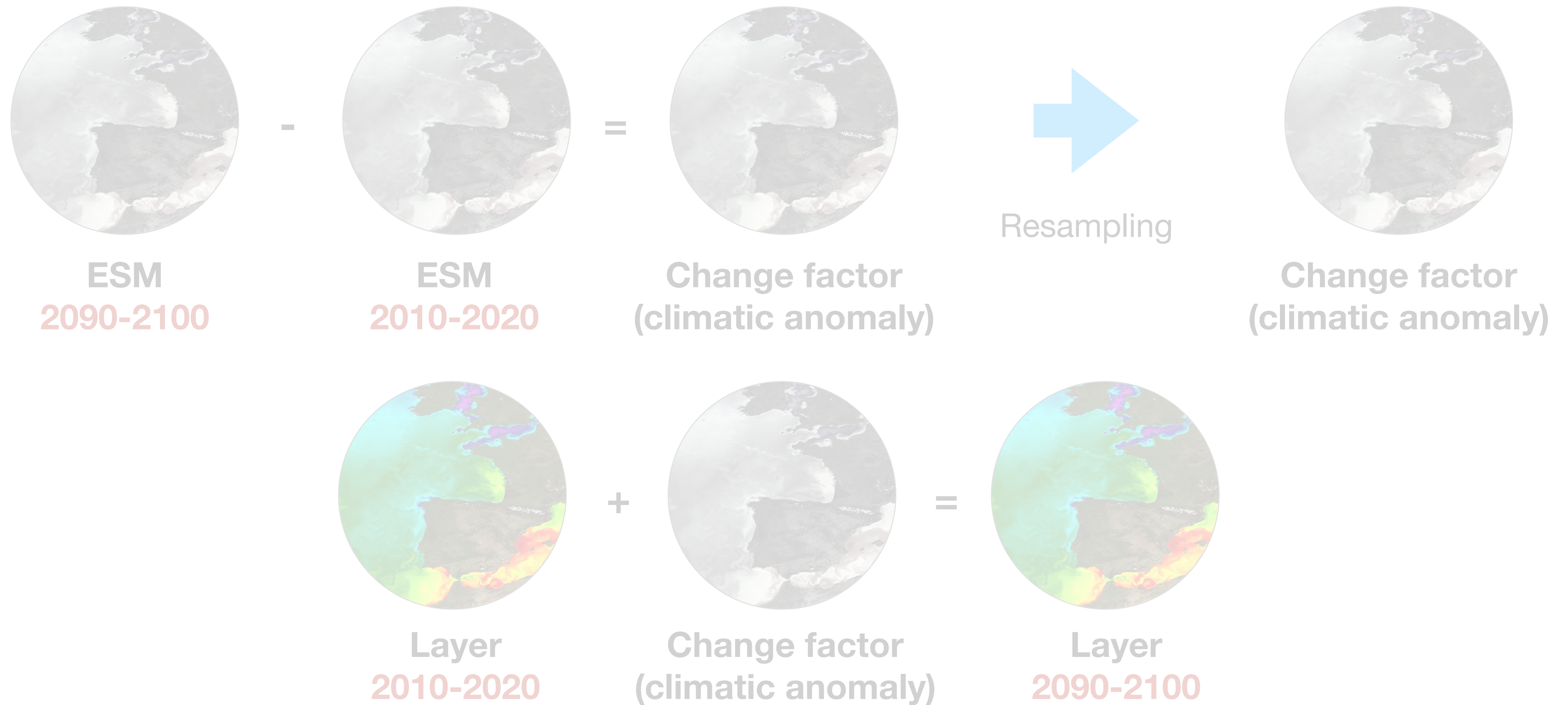
layer 1: Global maximum sea surface temperature of 2010-2020



Interpolation

Raw **data are collected** from numerous sources (e.g., buoys, weather stations) and their information are spatially interpolated to infer the distribution of the environmental gradients for the whole region - **estimates unknown points**.

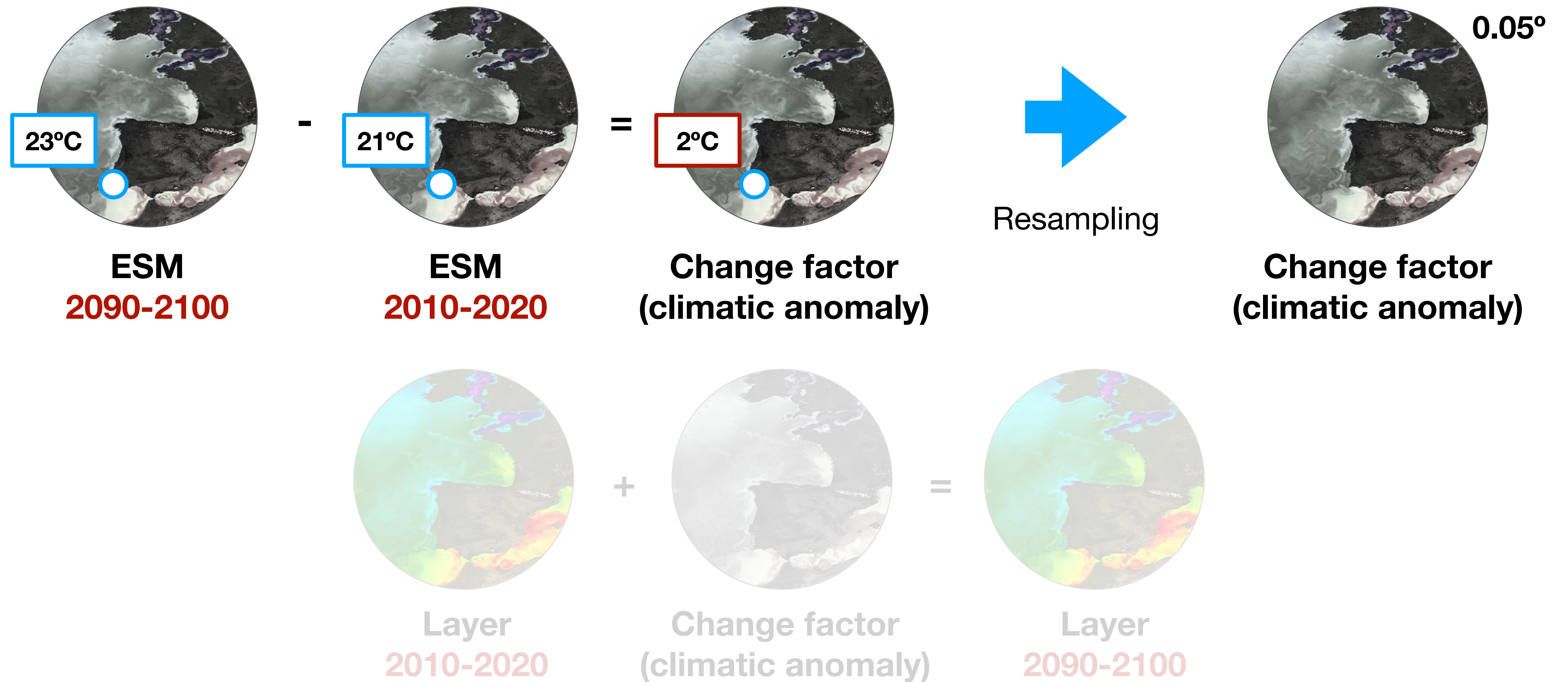
e.g., Inverse Distance Weighting algorithm.



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

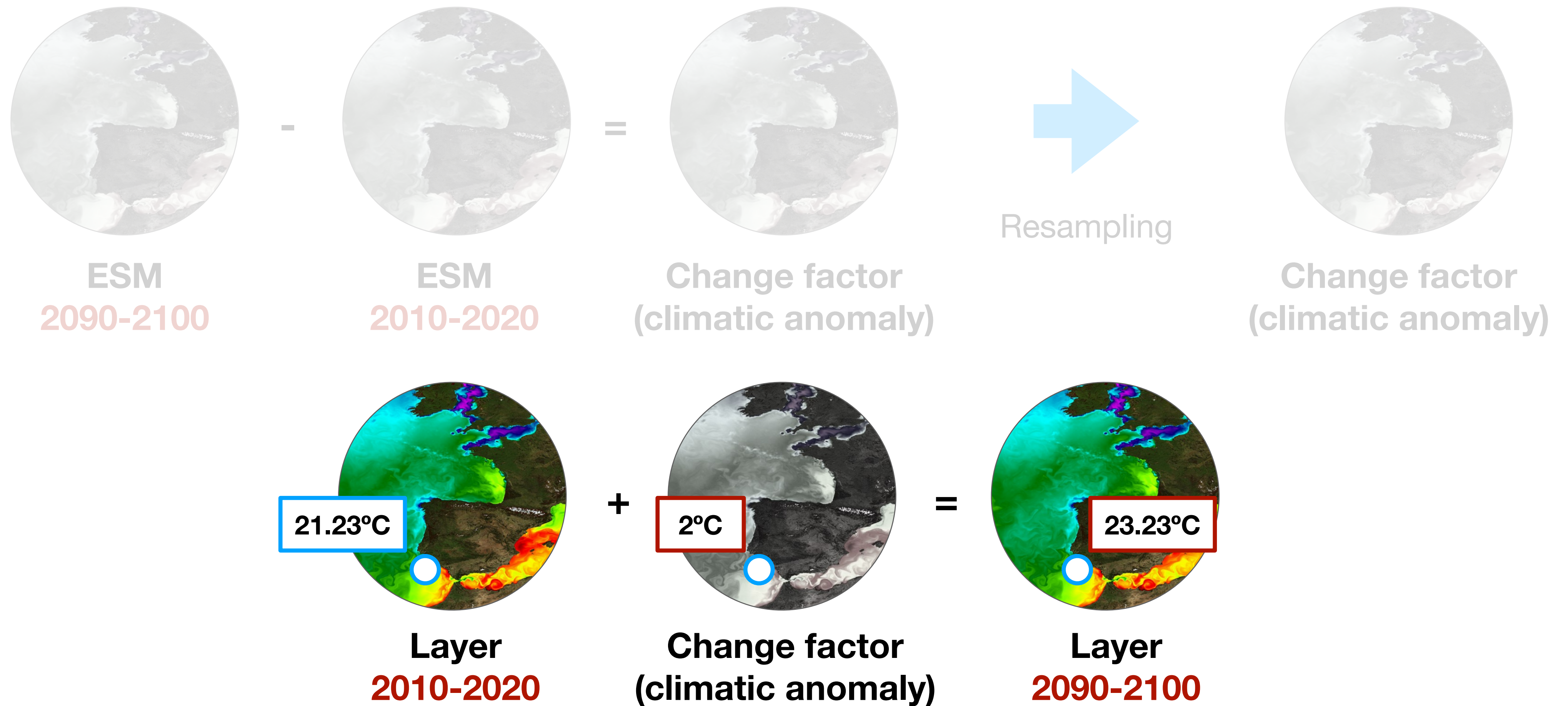
layer 2: Global maximum sea surface temperature of 2090-2100



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100

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- 19. Bio-ORACLE v2.0 Extending marine data layers for bioclimatic modelling
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